

Big **Insights** For Small Bakeries

Introduction

The Bakery App is an advanced software solution made specifically for bakery businesses, aimed at revolutionizing their daily operations. This comprehensive platform integrates essential functions such as order management, inventory tracking, and task scheduling into a unified system. By automating routine processes and providing real-time data, the app significantly enhances operational efficiency, reduces manual errors, and streamlines workflow.

With features designed to simplify complex tasks, the Bakery App enables bakery owners and staff to focus more on crafting high-quality baked goods and providing excellent customer service. It supports bakeries in maintaining optimal inventory levels, managing customer orders with ease, and ensuring that daily tasks are efficiently completed. Overall, the Bakery App is committed to helping bakeries improve their productivity, make informed decisions, and deliver an exceptional experience to their customers.

Project Overview

The Bakery App addresses the common challenges faced by bakeries, such as managing inventory, tracking sales, and organizing daily tasks. By integrating these functionalities into a single platform, the app offers a user-friendly experience that simplifies operations and improves overall business performance.

Why Bakeries

We chose bakeries in Germany as the focus of our project due to the increasing challenges they face in today's competitive market. Many traditional bakeries are struggling with poor management and lack of efficient planning, which have resulted in financial difficulties and closures across the country. Rising costs in ingredients and operations, along with the growing competition from large chains and supermarkets, have further strained these small businesses. With their deep cultural significance and long-standing tradition, the decline of local bakeries is a concerning trend that threatens the very fabric of German communities.

Key Features

The Bakery App is an innovative solution designed to enhance the operations of bakery businesses by streamlining the management of orders, inventory, and daily tasks. This comprehensive platform allows bakery staff to focus on creating quality products and delivering excellent customer service, thereby boosting efficiency and productivity.

The project began with the identification of key challenges faced by bakery operations, such as inventory management, sales tracking, and task organisation. In response to these challenges, the Bakery App was conceptualized to integrate these functionalities into a single, user-friendly interface. The development process started with low-fidelity wireframes, which served as visual blueprints outlining the basic structure and flow of the app. These prototypes were crucial in ensuring an intuitive and efficient user experience (UX).

The app offers several key features to address the needs of bakery businesses. One of the primary features is the dashboard, which provides an overview of daily activities, including sales, tasks, and inventory. This allows bakery managers to easily keep track of everything at a glance. Another essential feature is order management, which helps staff track and manage orders efficiently, ensuring that all orders are fulfilled promptly and accurately. Inventory management is also a critical component, with tools to monitor stock levels and receive alerts when inventory is running low. This feature ensures that the bakery is always well-stocked and prepared. Task management helps organize and prioritize daily tasks for the bakery staff, ensuring that all necessary tasks are completed on time. Lastly, sales analytics provide insights into sales data, enabling bakers to analyze trends and make informed business decisions to optimize their operations.

The UI/UX design of the Bakery App focuses on simplicity and functionality. The interface is clean and uses a monochromatic colour scheme to ensure readability and minimize distractions. Each screen is meticulously designed to provide maximum information without clutter, facilitating quick access to essential data. Consistency in design elements and interactions across all screens ensures a uniform user experience. Clarity is achieved through clear labelling and straightforward navigation, preventing user confusion. Efficiency is a core design principle, with fast access to features and data that improve workflow. Additionally, the app's layout is responsive, ensuring seamless operation on various devices, from tablets to smartphones.

Visual design elements further enhance the user experience. The typography is easy to read, improving legibility, while simple and intuitive icons and buttons facilitate quick actions. Charts and graphs provide visual representations of data, aiding in quick analysis and decision-making. These design principles ensure that the Bakery App is not only functional but also aesthetically pleasing and easy to use.

In conclusion, the Bakery App is a powerful tool that optimises bakery operations by integrating order management, inventory tracking, task organization, and sales analytics into a single platform. The thoughtful UI/UX design ensures a positive user experience, making the app an invaluable asset for bakery businesses aiming to enhance their efficiency and

productivity. By addressing the common challenges faced by bakeries, the Bakery App helps businesses streamline their operations and focus on what they do best—making great products.

Unstructured Interviews

Unstructured interviews played a pivotal role in shaping our app design and development process. By speaking directly with individuals in the baking industry, we were able to gather authentic insights into their day-to-day struggles, business operations, and pain points. These conversations, being open-ended and fluid, allowed the interviewees to express their thoughts freely, providing us with a deeper understanding of the challenges they face, such as poor management, rising costs, and intense competition. This raw, unfiltered feedback was instrumental in refining our ideas and steering our focus toward the most pressing problems that needed solutions.

The insights gained from these interviews helped us grow our initial concept into something much more robust and user-centered. As a result, our idea evolved to be more minimalistic in nature. Initially, we had envisioned a more feature-rich app, but through these discussions, we realized that simplicity and ease of use were paramount for bakers, who often have limited time and resources to dedicate to learning new technologies. The interviews fundamentally shifted our approach, guiding us to design an app that truly helps bakers by focusing on the essentials—streamlined inventory management, clear financial analysis, and simple product tracking.

These unstructured interviews not only influenced our UI/UX decisions but also justified many of our design choices. We moved toward a more minimal, clean design that wouldn't overwhelm users but instead offer practical and intuitive solutions. Ultimately, the feedback allowed us to validate and reshape our ideas, ensuring that our app would be a practical, functional, and valuable tool for bakers in addressing their real-world challenges.

UI/UX and Design Ideation

The Bakery App is designed to streamline bakery operations by integrating order management, inventory tracking, task organization, and sales analytics into a single, user-friendly platform. This app allows bakery staff to focus on producing quality products and providing excellent customer service.

The development process began with low-fidelity wireframes to outline the basic structure and flow of the app. These prototypes helped visualize the layout and functionality, ensuring an intuitive and efficient user experience (UX).

The app features a dashboard that provides an overview of daily activities, including sales, tasks, and inventory. Orders management allows staff to track and manage orders efficiently. Inventory management tools monitor stock levels and alert when inventory is low, ensuring the bakery is always prepared. Task management helps organize and prioritize daily tasks for the bakery staff. Sales analytics provide insights into sales data, enabling bakers to make informed business decisions.

The Bakery App includes several distinct screens: the welcome screen greets users and provides an overview of the day's expectations; the task management screen lists tasks to be completed; the check-in screen allows staff to check in for shifts and view inventory levels; the daily recap screen summarizes the day's sales and customer interactions; the orders screen displays current and future orders; the sales analytics screen provides an analysis of sales data over different periods with recommendations for improvement; and the inventory management screen lists inventory items with details on stock levels, prices, and alerts for low stock.

The UI/UX design focuses on simplicity and functionality. The interface is clean and easy to navigate, with a monochromatic color scheme that ensures readability and minimizes distractions. Each screen provides maximum information with minimal clutter, allowing users to quickly access the data they need. Key design principles include consistency, clarity, efficiency, and responsiveness.

Visual design elements enhance the user experience. Typography is easy to read, icons and buttons are simple and intuitive, and charts and graphs provide visual data representations for quick analysis and decision-making. By adhering to these design principles, the Bakery App ensures a positive user experience, making it an invaluable tool for optimizing bakery operations.

In summary, the Bakery App helps bakery businesses streamline their operations and focus on what they do best making great products.

The Journey

An app for bakeries in Germany to improve their operational efficiency was an enriching experience, marked by consistent progress and key learning moments. This journey, which included five critical checkpoints—idea pitch, low-fidelity wireframing, visual design high-fidelity, and final delivery—taught us valuable lessons about design thinking, collaboration, and product development.

#1 Idea pitch

From the very beginning, our project started on a positive note. The idea pitch set the tone for the entire process, as it highlighted the critical pain points in the bakery industry: poor planning, rising costs, and the resulting closures. Presenting this as a compelling argument validated our project's relevance and potential impact. This strong start gave us momentum and guided our research into building a strong MVP. We dove into research, exploring features that could make a difference in improving bakery operations. Our findings led us to focus on crucial functionalities such as data visualisation, financial analysis, smart inventory management, and product tracking—features that could streamline operations and address the core problems we identified early on.

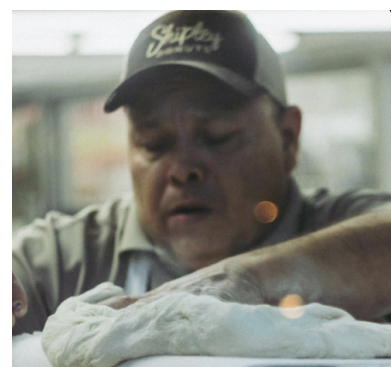
#2 Low-Fidelity

One of the turning points in our process was the creation of a storyboard. By visualizing how the app would be used in a real-world setting, we could see both the strengths and weaknesses of our initial ideas. This step gave us the insight to ensure that our app would not hinder daily operations but would instead assist bakeries in becoming more efficient. A key decision emerged from this stage: we needed to strike a balance between educational content and operational support, focusing on solving the issues of cost and competition. The development of a use-flow diagram was another significant milestone. This helped us outline key sections within the app and how to organize data effectively.

User-Centric Exploration

Borat Becker

Background: Borat Becker is a 52-year-old seasoned baker with over 30 years of experience in the bakery business.



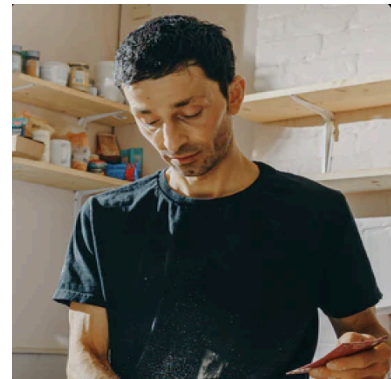
B A C K E R Y BACKBONE

Challenges: Despite his extensive experience, Borat struggles to direct his business toward the customers, facing difficulties in attracting and retaining them. His traditional approach may not align well with modern marketing and customer engagement strategies.

John Scott

Background: John Scott is a 25-year-old new founder in the bakery business, with no prior experience.

Challenges: John lacks experience in understanding what products to produce and sell. He needs to identify a niche or unique selling proposition (USP) to differentiate his bakery from competitors and is unfamiliar with effective marketing and business management strategies.



Low-Fidelity Prototypes

The development process started with low-fidelity wireframes to outline the basic structure and flow of the app. These prototypes helped in visualizing the layout and functionality, ensuring that the user experience (UX) would be intuitive and efficient.

The app offers several key features to address the needs of bakery businesses. One of the primary features is the dashboard, which provides an overview of daily activities, including sales, tasks, and inventory. This allows bakery managers to easily keep track of everything at a glance. Another essential feature is order management, which helps staff track and manage orders efficiently, ensuring that all orders are fulfilled promptly and accurately. Inventory management is also a critical component, with tools to monitor stock levels and receive alerts when inventory is running low. This feature ensures that the bakery is always well-stocked and prepared. Task management helps organize and prioritize daily tasks for the bakery staff, ensuring that all necessary tasks are completed on time. Lastly, sales analytics provide insights into sales data, enabling bakers to analyze trends and make informed business decisions to optimize their operations.

The Bakery App's design includes several defined screens, each tailored to specific aspects of bakery management. The welcome screen greets the user and provides an overview of the day's expectations, including the number of customers and revenue. The task management screen lists tasks to be completed, such as preparing goods and baking rolls, with options to add new tasks. Staff can check in for their shifts and view inventory levels on the check-in screen, which highlights items that are running low. The daily recap screen summarizes the day's sales and customer interactions, comparing actual figures to expected ones. The orders screen displays current and future orders, with details on preparation and delivery times. The sales analytics screen provides an analysis of sales data over different periods and offers recommendations for actions to improve performance. Finally, the

inventory management screen lists inventory items with details on stock levels, prices, and alerts for low stock.

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#3 Wireframing

As we progressed into the wireframing phase, it was here that we began to visualize the user experience. Our discussions around the visual design highlighted the need for a modern, dynamic interface—something that could move beyond a traditional POS system app to become an engaging and creative tool. This was a crucial moment where we realized that the app's aesthetics and interaction design were just as important as its functionality. We made the decision to introduce more dynamics making it slicker.

#4&5 Visual Design & High - Fidelity

As we moved into the high-fidelity design phase, our vision for the app solidified. We were deliberate in choosing a design system that balanced all elements, ensuring cohesion in

iconography, fonts, and colours. One of the standout features we incorporated was a dark and light mode, which added to the app's adaptability by accommodating different user environments and preferences. These design choices reflected our commitment to creating an intuitive, attractive, and user-friendly product.

#6 Final Delivery

Overall, this project journey was both smooth and engaging. At each stage, we received valuable feedback that pushed us to think creatively and challenged us to find solutions to potential problems. Throughout the process, we gained a deeper understanding of design systems, user flows, and product development. This, in turn, empowered us to create a fully operational prototype that stands as a testament to our work.

In the end, we take pride in the strong, concrete ideas that emerged throughout the development process. Our prototype not only addresses the operational challenges bakeries face but also offers a modern, visually appealing, and engaging user experience. This project taught us that the key to successful app development lies in striking a balance between functionality, creativity, and user-centered design.

We walk away from this experience with a refined set of skills, a deeper understanding of the app development process, and confidence in our ability to turn strong ideas into tangible products.

Challenges and Learnings

The overall takeaway for us over the course of the semester has been how the process behind a successful User Interface Design is structured. The many steps between our initial idea and the finished app prototype taught us a lot about why each of the different stages is important on its own and how they are connected. On this learning journey as a team, we had many positive experiences but also faced some challenges that had to be overcome.

What went particularly well for us was the idea finding and preparation of the idea pitch. From the get-go, we felt very confident in our concept because we all recognized the significance of the bakery topic and what set of values we wanted to transport with our end goal in mind.

Furthermore, the presentations were a positive point for our team. We put a lot of effort into the preparation and always aimed to keep the audience engaged as much as possible by placing a high value on storytelling.

Speaking of, in other stages of the project, we could have improved our storytelling. This learning originated from having some struggles of finding the most important common ground between the two user journeys of Borat Becker and John Scott, leading to difficulties in deciding which features we would need to prioritize later down the road. To solve this in future projects and become more efficient, we could start by putting a higher focus on the empathy map and derive even more detailed user journeys from it.

In addition, coming up with the unstructured interviews as our way of evaluation took us some time and effort because it looked like a rather inefficient way of gathering useful data at first. Since most of the other approaches seemed to have no real use for our app concept at this point in the development process though, we decided on going with it anyway. Later, those interviews should prove to be a deciding factor in shaping our final app design and the scope of required features.

One of our toughest challenges was finding the fine line between the more classy aspects that one associates with small bakeries and the simplistic yet modern design approach that is often seen in applications related to finance and management because we wanted our design to embody both characteristics. That's why in the end, we decided on our mostly black and white color scheme with green aspects due to it communicating modernity as well as sustainability and growth and combined it with colourful yet simple illustrations of pastries.

A possible further improvement for our prototype could be a profile menu for general settings of the app, such as language, and making easy changes to selections made during the onboarding, like the bakery's cash register. This would provide more flexibility to the baker's which would be important in the long-term.